



Capital University of Science & Technology

Summer Internship Report

Name: Ayesha Babar

Reg no: BSE231036

Internship Duration: 6 weeks



DEPARTMENT OF SOFTWARE ENGINEERING

CAPITAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

1. Document Control

Item	Details
Report Title	Internship Progress Report
Student	Ayesha Babar
Registration No.	BSE231036
Program	BS Software Engineering
Department	Department of Software Engineering
University	Capital University of Science and Technology (CUST)
Organization	Yuni Pakistan
Internship Position	Tech Team Intern (Web Development)
Internship Start Date	03 August 2026
Internship Duration	Six weeks
Internship Course	SE4100 - Internship

Table of Contents

1. Document Control.....	2
2. Executive Summary.....	4
3. Acknowledgement.....	4
4. Introduction.....	5
4.1 Internship Overview.....	5
4.2 Internship Objectives.....	5
4.3 Role and Responsibilities.....	5
5. Organization Overview.....	6
5.1 Yuni Pakistan.....	6
5.2 Internship Environment and Responsibilities.....	6
6. Technical Areas and Tools.....	6
7. Detailed Internship Progress.....	7
7.1 Week 1 - Yuni Pakistan Website Enhancement.....	7
7.2 Week 2 - Landing Page and Google Apps Script Integration.....	8
7.3 Weeks 3-6 - WearVibe E-Commerce Application.....	9
7.4 Testing, Optimization, and Documentation.....	14
8. Learning Outcomes and Skills Developed.....	14
8.1 Technical Skills.....	14
8.2 Software Engineering Skills.....	14
8.3 Professional Skills.....	15
9. Challenges and Problem-Solving.....	15
10. Professional and Workplace Development.....	16
11. Conclusion.....	16
12. Offer Letter.....	17
13. Completion Certificate.....	20
14. Evaluation Form.....	21

2. Executive Summary

This internship progress report presents the work assigned during a six-week Web Development internship at Yuni Pakistan. The internship commenced on 03 August 2026 and was completed as a six-week period. The role was formally designated as "Tech Team Intern (Web Development)" and involved website development and maintenance, technical troubleshooting, software and automation-related work, digital tool optimization, implementation of technology solutions, and documentation.

The internship work was organized into three major streams. The first stream involved enhancement of the existing Yuni Pakistan website, including skeleton loaders, responsive navigation improvements, scroll animations, and a revised light and vibrant visual theme. The second stream involved development of a landing page and a Google Apps Script-based form integration for automated spreadsheet data storage. The third and largest stream involved full-stack development work for WearVibe a fully fledged e-commerce web application for curated matching clothes using React.js and Firebase backend services.

The WearVibe assignment covered user onboarding and personalization, a Discover landing page, a three-step Box Builder, matching rules, live outfit preview, price calculation, persistent cart functionality, checkout, authentication, user profiles, role-based administration, Cloud Firestore, Firebase Storage, and security rules. The final stage focused on functional testing, responsive testing, integration, troubleshooting, optimization, and documentation.

Overall, the internship provided practical exposure to real-world web development workflows, frontend and backend integration, cloud-based application services, authentication and authorization, testing, troubleshooting, and professional software development practices.

3. Acknowledgement

I would like to express my sincere gratitude to my supervisor, Abdul Moiz (Chief Technology Officer), and the technical team at Yuni Pakistan for providing me with the opportunity to undertake a professional internship in Web Development. I am deeply grateful to him for his

valuable guidance, technical oversight, and continuous feedback throughout my work on our development projects. His mentorship provided me with practical exposure to software development workflows, cloud database architectures, and best practices in a professional engineering environment.

I am also thankful to the Department of Software Engineering at Capital University of Science and Technology (CUST) for providing the academic framework and internship program that enabled me to connect my university learning with practical industry work. I appreciate the guidance, evaluation, and support provided throughout the internship process.

Finally, I acknowledge the contribution of all team members who provided technical guidance, constructive feedback, and coordination during the internship.

4. Introduction

4.1 Internship Overview

The internship was undertaken as part of the SE4100 Internship course in the Department of Software Engineering at Capital University of Science and Technology (CUST). The internship offered practical experience in Web Development and was carried out at Yuni Pakistan.

According to the internship offer letter, the assigned position was Tech Team Intern (Web Development), with a duration of six weeks starting from 03 August 2026. The stated responsibilities included assisting in website development, maintenance and updates; supporting technical troubleshooting; working on software, system, or automation-related projects; managing and optimizing digital tools; researching and implementing technology solutions; coordinating for technical support; completing assigned work on time; and maintaining documentation.

4.2 Internship Objectives

- Apply software engineering concepts in a practical professional environment.
- Gain hands-on experience in web application development and user interface enhancement.
- Understand frontend-to-backend integration through real project assignments.
- Develop practical experience with cloud-based backend services and user authentication.
- Improve debugging, testing, problem-solving, and documentation skills.
- Understand professional development practices, task ownership, communication, and timely delivery.

4.3 Role and Responsibilities

- Contribute to assigned website development and enhancement tasks.

- Implement interface improvements and responsive behavior.
- Develop and integrate forms and automation workflows.
- Support application features involving authentication, data management, storage, and administrative controls.
- Perform testing and troubleshooting of implemented functionality.
- Maintain clear documentation of assigned development work.

5. Organization Overview

5.1 Yuni Pakistan

Yuni Pakistan provided the internship placement for the six-week Web Development role. The internship offer letter describes the organization as Yuni (SMC-PVT) Limited and identifies the internship as part of its Tech Team. The letter states that the organization’s vision is inspired by the philosophy of Khudi and focuses on empowering students with tools, knowledge, and opportunities to create meaningful impact.

Within the internship role, the Tech Team environment provided exposure to digital platforms, website development, software and automation projects, technical support, and implementation of technology solutions.

5.2 Internship Environment and Responsibilities

The internship offer letter specifies a hybrid working arrangement in which working hours are assigned according to tasks, meetings, and project requirements. It also emphasizes attendance, active participation, professional communication, task completion, follow-up, team coordination, and professional growth. These requirements provided a structured context for completing technical assignments and maintaining consistent progress.

6. Technical Areas and Tools

Area	Tools / Services	Application During Internship
Web Development	Web application source code and responsive UI techniques	Website enhancement, landing pages, and WearVibe interface development
Frontend Integration	Interactive web components and UI behavior	Navigation, loaders, animations, product selection, cart, and checkout
Automation / Scripting	Google Apps Script	Form processing and automated spreadsheet data storage
Authentication	Firebase Authentication	Registration, login, logout, and password reset
Database	Cloud Firestore	Users, products, rules, carts, and orders
File Storage	Firebase Storage	Product image storage
Security	Firebase Security Rules	Data protection and role-based administrative access
Testing	Functional, responsive, and integration testing	Verification of application features and workflows

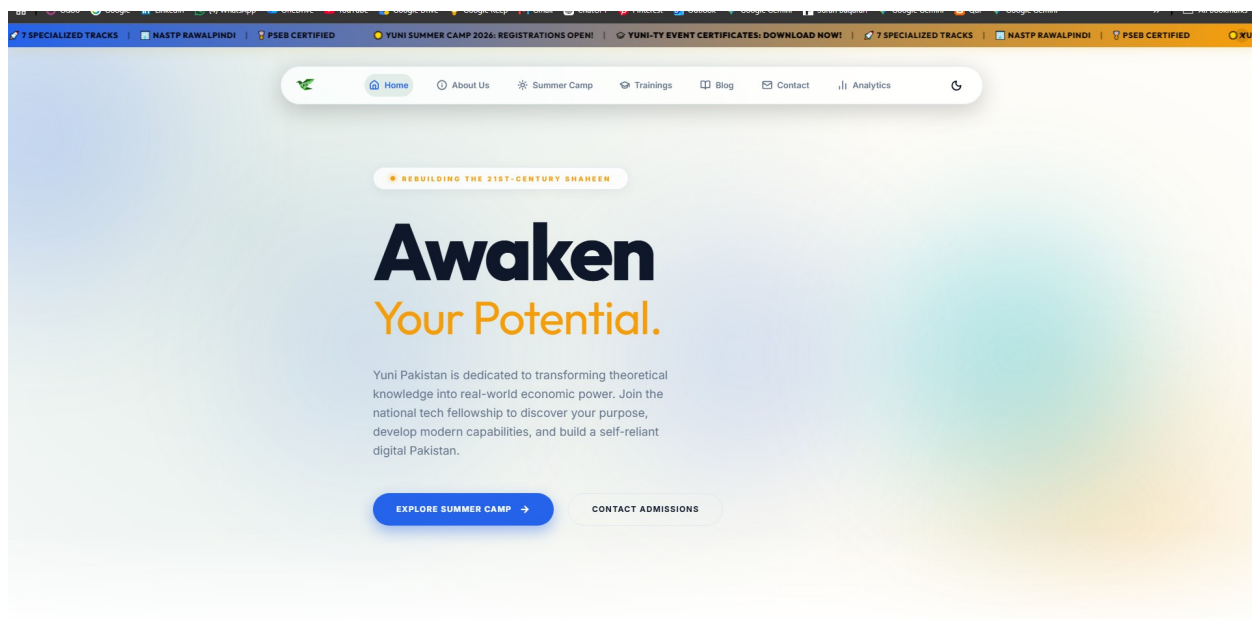
7. Detailed Internship Progress

7.1 Week 1 - Yuni Pakistan Website Enhancement

The first week focused on enhancement of the existing Yuni Pakistan website. The assignment was primarily concerned with improving the interface, loading experience, responsiveness, and visual presentation without changing the overall UI direction.

- Audited and reviewed the existing Yuni Pakistan website source code to identify layout and performance bottlenecks.
- Implemented reusable CSS skeleton loaders for website sections to provide visual loading feedback and reduce perceived latency.
- Integrated lightweight scroll-triggered animations to improve user interaction and visual flow without degrading page performance.
- Updated the website color scheme to a vibrant and light palette while strictly maintaining the existing UI structure.
- Explored suitable lightweight web animation resources and applied smooth CSS keyframe transition techniques.
- Tested the updated interface across multiple screen viewports to ensure consistent responsive rendering.

The first week established practical experience in working with an existing codebase rather than developing an interface from the beginning. It also introduced the importance of preserving existing design structure while making targeted improvements.



7.2 Week 2 - Landing Page and Google Apps Script Integration

The second week focused on developing a responsive landing page and establishing an automated connection between a data collection form and a spreadsheet through Google Apps Script. The central objective was to connect the frontend form with a backend process so that submitted data could be stored automatically.

- Developed a responsive, mobile-friendly landing page according to the provided requirements.
- Designed and implemented the interactive user registration and data collection form.
- Connected the form with Google Apps Script using serverless `doPost` handling for backend processing.
- Configured automated data routing to store submitted user records directly as rows in the designated cloud spreadsheet.
- Established the client-to-backend data connection using asynchronous JavaScript `fetch()` API calls.
- Conducted end-to-end testing to verify form submission sanitization and reliable spreadsheet storage.
- Identified and resolved edge-case issues involving submission timeouts, data payload handling, and cross-origin integration.

This assignment provided practical experience in lightweight backend automation and demonstrated how a web form can be integrated with a cloud-based scripting service and structured data storage workflow.

Adoption of Fintech Apps Among University Students in Pakistan

This survey is part of a university research project for a Probability & Statistics course. Your responses are anonymous and will be used solely for academic analysis. It takes approximately 5-7 minutes to complete.

* Required

Section A --- Demographic Information

1. What is your gender? * Nominal

☐ Male

☐ Female

☐ Prefer not to say

2. What is your age (in years)? * Ratio

Your answer

3. What is your field of study? * Nominal

☐ Business / Management

☐ Engineering

☐ Computer Science / IT

☐ Social Sciences

☐ Other

4. What is your monthly personal income or allowance (PKR)? * Ratio

Your answer

Section B --- Fintech App Usage

5. Which fintech app do you use most frequently? * Nominal

☐ EasyPaisa

☐ JazzCash

☐ SadaPay

☐ NewPay

```

1  const SHEET_NAME = 'Responses';
2
3  const HEADERS = [
4    'Timestamp',
5    'university',
6    'province',
7    'age_group',
8    'gender',
9    'degree_level',
10   'monthly_spending',
11   'apps',
12   'usage_frequency',
13   'purposes',
14   'ease_of_use',
15   'security',
16   'trust',
17   'time_saving',
18   'affordability',
19   'recommendation',
20   'problems',
21   'prefer_apps',
22   'future_use',
23   'improvement',
24   'comments'
25 ];
26
27 function doPost(e) {
28   try {
29     const spreadsheet = SpreadsheetApp.getActiveSpreadsheet();
30
31     const sheet =
32       spreadsheet.getSheetByName(SHEET_NAME) ||
33       spreadsheet.insertSheet(SHEET_NAME);
34
35     if (sheet.getLastRow() === 0) {
36       sheet
37         .getRange(1, 1, 1, HEADERS.length)
38         .setValues([HEADERS]);
39     }
40
41     const data = JSON.parse(e.postData.contents || '{}');
42
43     const row = HEADERS.map((header, index) => {
44       return index === 0 ? new Date() : data[header] || '';
45     });
46   }
47 }

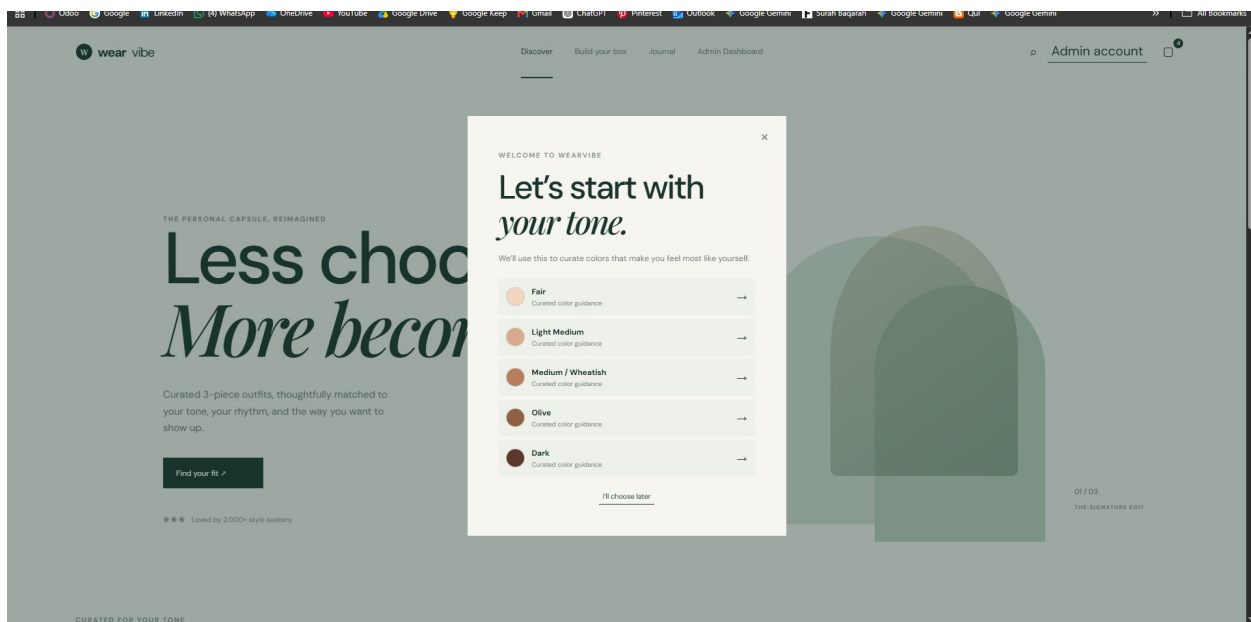
```

7.3 Weeks 3-6 - WearVibe E-Commerce Application

From Week 3 onward, the principal assignment was the development of WearVibe, a premium and responsive e-commerce web application. The project was planned as a complete web application with personalized onboarding, product discovery, product configuration, cart and checkout features, user authentication, administrative management, and Firebase-based backend services.

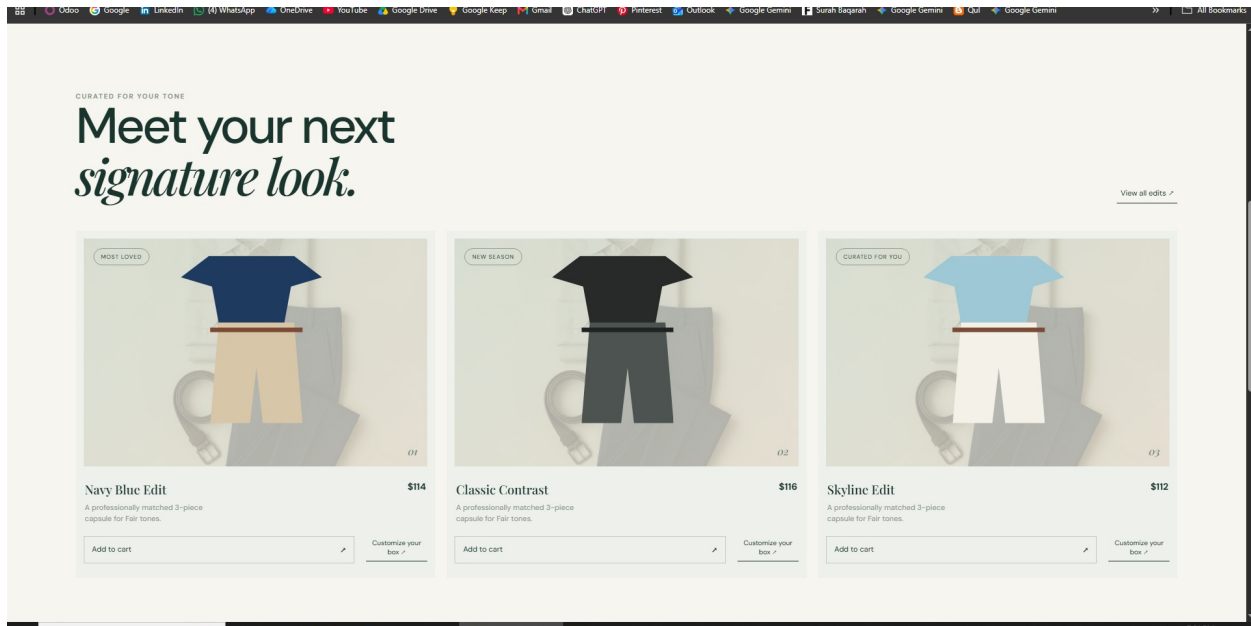
User Onboarding and Personalization

- Implemented a guided welcome questionnaire to collect the user's skin tone and style preferences.
- Programmed recommendation logic to provide personalized color guidance based on selected skin tone.
- Connected onboarding choices to display curated outfit suggestions tailored to the user profile.



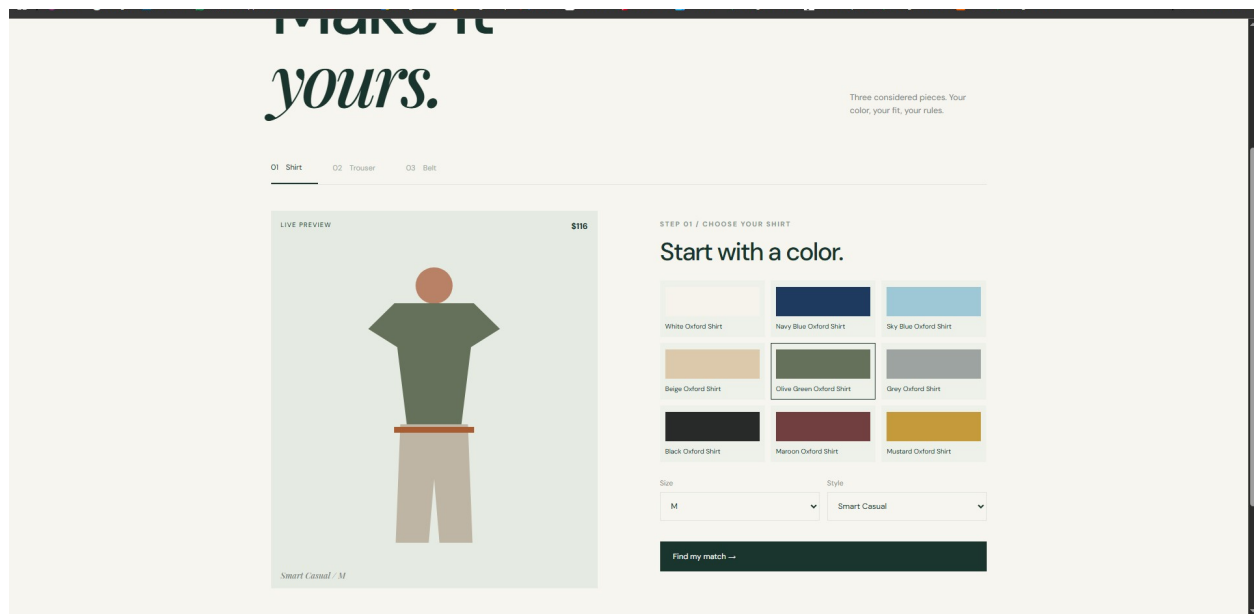
Storefront and Product Discovery

- Developed the Discover storefront page showcasing pre-curated signature capsule edits.
- Structured and presented Shirts, Trousers, and Belts as core browsable product categories.
- Engineered a responsive card grid interface providing a modern storefront experience.



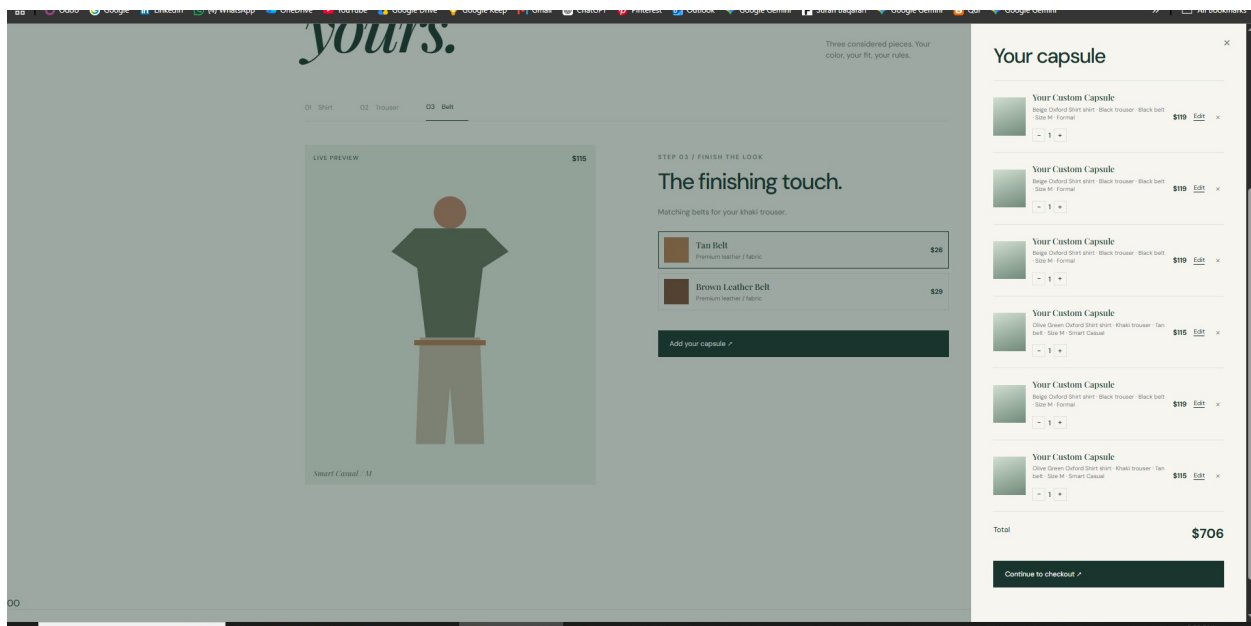
Box Builder

- Developed a sequential three-step Box Builder for Shirt, matching Trouser, and complementing Belt selection.
- Implemented client-side matching algorithms between shirts and trousers to guide color coordination.
- Built an interactive preview canvas delivering live visual feedback of the chosen 3-piece outfit.
- Programmed the preview figure to update colors and styling layers dynamically upon user selection.
- Created dynamic price calculation logic to compute package bundle pricing and discounts in real time.



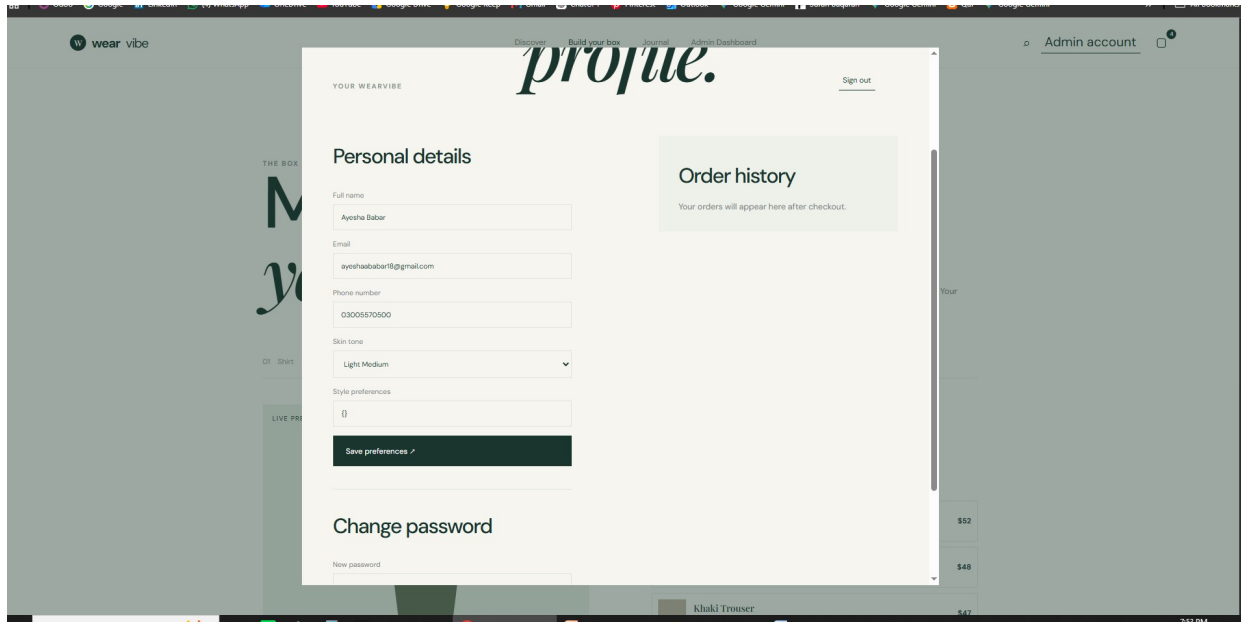
Cart and Checkout

- Engineered a persistent shopping cart supporting browser local storage for guests and cloud syncing for users.
- Implemented an animated slide-out cart drawer for rapid cart inspection.
- Developed in-cart item configuration editing, quantity adjustments, and item removal.
- Developed the checkout flow with input validation for Name, Phone, Shipping Address, and City.
- Integrated payment options including Cash on Delivery and Direct Bank Transfer with confirmation prompts.



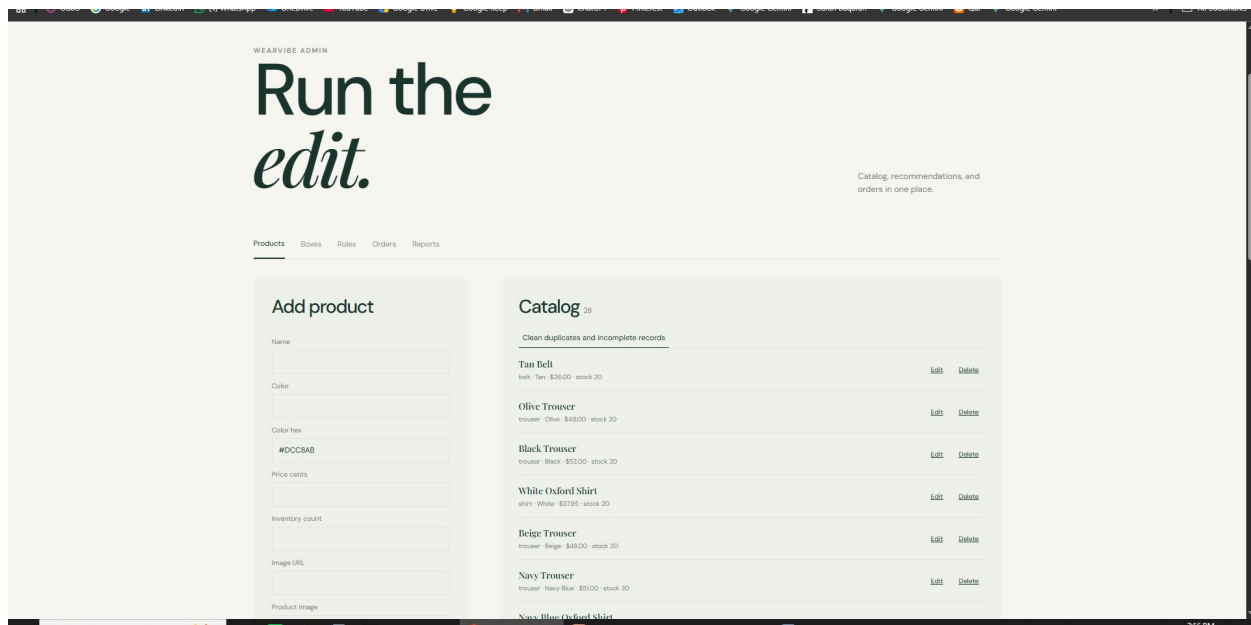
User Accounts

- Integrated Firebase Authentication for email/password user registration, login, logout, and password recovery.
- Built secure authentication modals (`AuthPanel.jsx`) with real-time error handling.
- Developed a user profile dashboard (`ProfilePanel.jsx`) for managing account details and style preferences.



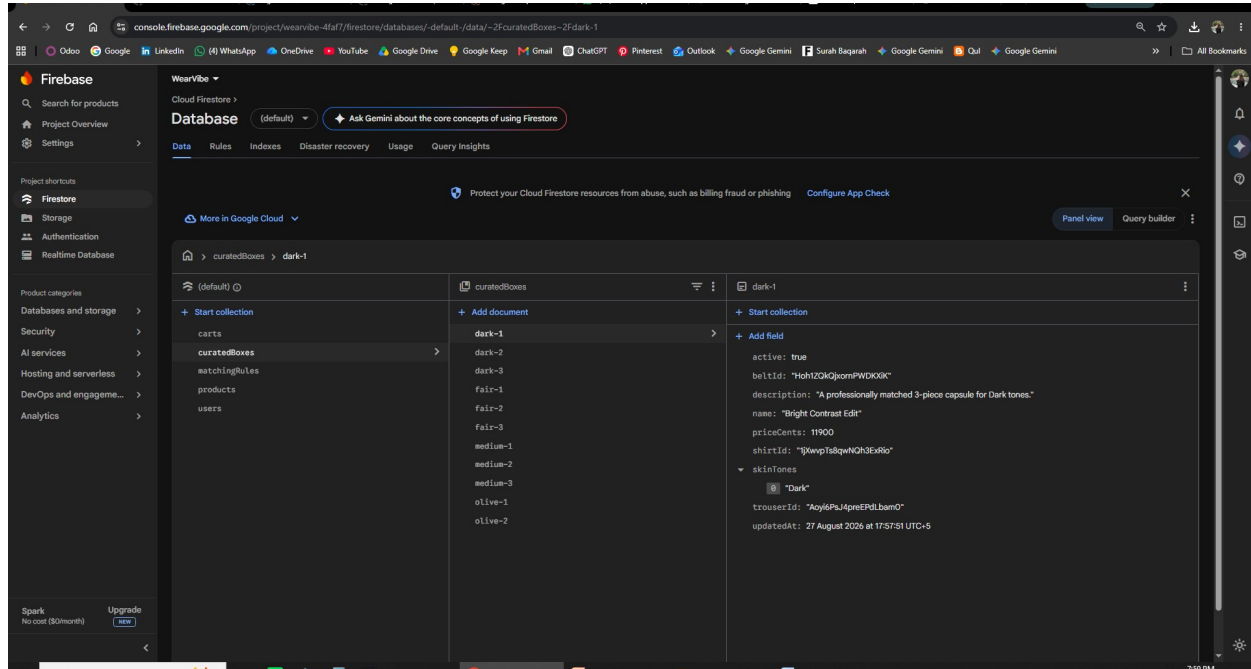
Admin Management

- Developed a role-based administrative dashboard interface (`AdminPanel.jsx` , `BoxManager.jsx`).
- Implemented administrative tools to add, edit, and manage catalog inventory for Shirts, Trousers, and Belts.
- Configured admin features for updating product matching rules and assembling curated boxes.
- Created role validation utilities (`checkAdmin.mjs`) to restrict admin endpoints to authorized users.



Backend and Security

- Configured Cloud Firestore NoSQL schemas for users, products, rules, carts, and customer orders.
- Configured Firebase Storage buckets for high-resolution product imagery.
- Authored and deployed strict Firebase security rules (` firestore.rules` , ` storage.rules`) to protect user data and secure catalog writes.



7.4 Testing, Optimization, and Documentation

- Performed functional testing across all implemented frontend and backend features.
- Tested the complete WearVibe workflow from onboarding questionnaire through final checkout submission.
- Tested Firebase Authentication sessions, persistent login states, and profile updates.
- Verified shopping cart state management, checkout validation, and payment method selection.
- Tested administrative capabilities and verified role-based access restrictions.
- Verified Cloud Firestore read/write operations, real-time snapshot listeners, and Storage integration.
- Audited and tested Cloud Firestore security rules against unauthorized document access.
- Performed responsive testing across diverse screen viewports and mobile devices using Chrome DevTools.
- Identified and resolved UI layout glitches, touch interaction defects, and backend integration edge cases.
- Optimized application rendering performance and overall user experience.
- Completed final codebase cleanup, code comments, and technical project documentation.

8. Learning Outcomes and Skills Developed

8.1 Technical Skills

- Enhanced practical understanding of web application development and maintenance.
- Improved ability to work with an existing website codebase and perform controlled UI enhancements.
- Experience with responsive design and mobile-oriented interface optimization.
- Practical understanding of asynchronous loading feedback through skeleton loaders.
- Experience implementing interactive scroll-based UI behavior.
- Practical experience integrating frontend forms with Google Apps Script.
- Understanding of Firebase Authentication for user account management.
- Understanding of Cloud Firestore as a cloud database for application data.
- Understanding of Firebase Storage for image and file management.
- Experience with security rules and role-based access control concepts.
- Experience testing complete application workflows and troubleshooting integration issues.

8.2 Software Engineering Skills

- Requirement interpretation and feature decomposition.
- Incremental implementation of application modules.
- Debugging and root-cause-oriented problem solving.
- Functional, integration, and responsive testing.
- Technical documentation and structured reporting.
- Maintaining consistency between user interface requirements and backend functionality.

8.3 Professional Skills

- Task planning and prioritization.
- Working within assigned timelines and deliverables.
- Professional communication and follow-up.
- Coordination with technical and organizational stakeholders.
- Adaptability when moving between different project types and technologies.
- Maintaining organized records of development progress.

9. Challenges and Problem-Solving

Working with an Existing Codebase

The first assignment required improvements to an existing website rather than a new implementation. The approach was to understand the existing structure first and then make targeted changes while maintaining the original interface direction.

Responsive Behavior

Improving mobile navigation and ensuring that visual enhancements remained usable on different screen sizes required repeated responsive checks and refinement.

Frontend-to-Backend Integration

The landing-page assignment required a reliable connection between the user-facing form, Google Apps Script, and spreadsheet storage. Testing the complete submission path helped identify and resolve integration issues.

Complex Feature Integration

WearVibe combined personalization, product matching, live preview, dynamic pricing, cart management, checkout, authentication, and administration. Breaking the project into feature groups reduced complexity and made testing more manageable.

Access Control and Security

The application required different capabilities for regular users and administrators. Firebase authentication and security rules were therefore treated as a separate implementation and testing concern.

Final Validation

The final phase required testing several interconnected workflows rather than individual features in isolation. End-to-end testing and responsive verification were used to identify issues that could appear only after multiple features were combined.

10. Professional and Workplace Development

The internship provided experience beyond technical implementation. The tasks required interpreting written requirements, organizing work into manageable components, following project instructions, testing before completion, and maintaining clear documentation.

- Improved confidence in handling assigned development tasks independently.
- Developed greater awareness of the importance of testing before delivery.
- Improved understanding of how frontend requirements connect to backend services.
- Strengthened professional communication, task follow-up, and documentation habits.
- Developed a stronger appreciation for maintainability, security, and responsive behavior in web applications.

11. Conclusion

The six-week internship at Yuni Pakistan provided a valuable practical learning experience in Web Development. The work progressed from enhancing an existing website to implementing a form-based automation workflow and then to developing a multi-feature web application with cloud backend services.

The first two assignments developed practical skills in interface enhancement, responsive design, animation, loading states, form development, and backend automation. The WearVibe project extended this experience into a broader software engineering workflow involving user onboarding, product configuration, cart and checkout functionality, authentication, administrative management, database integration, storage, and security.

The internship also strengthened testing, debugging, documentation, task management, and professional communication skills. Overall, the experience helped bridge the gap between academic software engineering concepts and practical web application development in a professional environment.



Ayesha Babar
Student / Intern
Reg No: BSE231036
Department of Software Engineering, CUST



Abdul Moiz
Chief Technology Officer (CTO)
Yuni (SMC-Pvt) Limited, NASTP
Official Stamp & Signature

12. Offer Letter



INTERNSHIP OFFER LETTER

Date: 28 July 2026

Dear Ayesha Babar,

Position: Tech Team Intern (Web Development)

We are pleased to offer you the position of Tech Team Intern at Yuni Pakistan for a period of 6 weeks, commencing from 3 August, 2026. Your selection reflects our confidence in your potential to contribute towards our mission of rebuilding the 21st-century Shaheen through innovation, learning, and practical skill development.

At Yuni Pakistan, our vision is inspired by the philosophy of Khudi and centers on empowering students with the tools, knowledge, and opportunities to create meaningful impact. As part of our Tech Team, you will play an important role in strengthening the digital ecosystem and supporting our both technological initiatives.

Your responsibilities during this internship will include, but are not limited to:

- Assisting in website development, maintenance, and updates.
- Supporting technical troubleshooting and resolving digital platform issues.
- Working on software, system, or automation-related projects assigned by the team lead.
- Managing and optimizing digital tools used within Yuni's ecosystem.
- Assisting in research and implementation of innovative tech solutions for organizational growth.
- Coordinating with other departments to provide technical support where required.
- Ensuring timely completion of assigned tasks and maintaining proper documentation of work.

Internship HR Policies and Terms

Working Hours (Hybrid)

Internship hours will be assigned based on tasks, meetings, and project requirements. Flexibility may be required depending on operational needs.

Attendance Policy

Regular attendance and active participation are mandatory. Interns are expected to remain responsive and available for assigned tasks.

Professional Conduct

Interns must maintain professionalism, discipline, and respectful communication while representing Yuni Pakistan.

Confidentiality

All internal documents, partnership details, and organizational discussions must remain confidential during and after the internship.

Learning and Performance

This internship is performance-based and intended for skill-building. Your performance will be reviewed periodically based on:

- Communication quality
- Task completion
- Follow-up consistency
- Team coordination
- Professional growth

Three Strike Rule



**YUNI
PAKISTAN**

NASTP

- YUNI (SMC-PVT) LIMITED
- CEO@YUNIPAKISTAN.COM
- [+92 321 561 5401](tel:+923215615401)

Yuni Pakistan follows a structured disciplinary policy:

Strike One: Verbal warning

Strike Two: Written warning

Strike Three: Internship termination

Strikes may be issued for:

- Missing deadlines
- Failure to follow up on assigned leads
- Unapproved absence
- Unprofessional communication
- Negligence in assigned responsibilities
- Repeated lack of engagement

Completion Certificate

Interns who successfully complete their duration and meet performance standards will be awarded an Internship Completion Certificate and may be considered for future permanent roles.

Termination Policy

Either party may terminate this internship with 3 days prior notice, unless termination occurs due to serious misconduct or violation of company policies.

By signing below, you acknowledge and accept the responsibilities and policies outlined in this letter.

We look forward to your contribution and hope this experience helps in your professional development.

Intern Name: _Ayesha_Babar_

Date: _28 / 7 / 2026_

Authorized By:

CTO & HR Department

Yuni Pakistan



13. Completion Certificate



14. Evaluation Form



Capital University of Science and Technology

Expressway, Kahuta Road, Zone-V, Islamabad
Phone: 92-51-111-87-87-87 Fax: 92-51-4466705
Email: info@jinnah.edu.pk Website: http://www.jinnah.edu.pk

Internship Evaluation Form

The purpose of this evaluation form is to solicit your opinion about the performance of student during this course of Internship at your organization. We appreciate your honest and objective response.

Student's Name: Ayesha Babar
Registration Number: BSE231036
Organization/Company of Internship: Yuni Pakistan
Duration of Internship in weeks: 6 weeks

Please evaluate the student's performance and conduct during Internship Training by encircling the appropriate number on the scale of 0 to 10. (0 being Lowest rating and 10 being Highest)

- Professional knowledge and skill.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Ability of student to apply his/her knowledge.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Analytical approach towards engineering problem solving.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Capability to adapt to a new working environment.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Initiative and drive.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Ability to meet the given target in specified time duration.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Self-reliance in accomplishing the task.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Ability to cooperate with associates.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Punctuality and discipline.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----
- Interpersonal skills.

0	1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	---	----

Project/Task assigned: Landing Pages, Website Enhancement, Fully fledged e-commerce website for matching clothes

Percentage of assigned Project/Task completed: ✓

0 to 20 %	20 to 40 %	40 to 60 %	60 to 80 %	80 to 100 %
-----------	------------	------------	------------	-------------

Remarks: The internee was well equipped with real world knowledge, they succeeded in completing all the tasks and projects that were assigned to them on time. They have a good tendency of shaping according to their environment

Abdul Moiz (CTO) +966 53 396 6140
Evaluator's Name, Designation and Contact Information

Signature
Signature, Stamp and Date

